

Part II

This panel dataset contains economic indicators for the United States, Australia, Brazil, Canada, China, Egypt, France, Germany, India, Indonesia, Japan, Mexico, South Africa, United Kingdom, and Vietnam from 2004 to 2023, including:

GDP Growth (%) – Measures economic growth

Unemployment Rate (%) – Percentage of unemployed in the labor force

Inflation (%) – Annual price level changes

Trade Openness (% of GDP) – Sum of exports and imports as a share of GDP

Each observation represents a country-year, making it suitable for panel data analysis.

Research Question: How does GDP growth affect unemployment across countries?

According to Okun's Law, GDP growth and unemployment typically have an inverse relationship, but the strength of this effect may vary by country.

We hypothesize that higher GDP growth will result in lower unemployment. Other variables like inflation and trade openness could influence unemployment as well.

Table 1: The output below summarizes the statistics of key economic variables amongst the countries in the data set. From 2004 to 2023, GDP growth rates range from -10.297% to 14.231%. Unemployment rates had significant variance, ranging from 0.999% to 34.007%. Inflation peaked at 33.885%, and trade openness as a percentage of GDP ranged from 22.11% to 186.68%.

```
summary(df) # Provides mean, min, max, and quartiles for each variable
```

```
##      ...1      Country      Country_Code      Year
## Min.    : 0.00  Length:300  Length:300      Min.    :2004
## 1st Qu.: 74.75  Class :character  Class :character  1st Qu.:2009
## Median :149.50  Mode  :character  Mode  :character  Median :2014
## Mean    :154.83                                Mean    :2014
## 3rd Qu.:244.25                                3rd Qu.:2018
## Max.    :319.00                                Max.    :2023
##      GDP_Growth      Inflation      Trade_Openness      Unemployment
## Min.    :-10.297  Min.    :-1.353  Min.    : 22.11  Min.    : 0.999
## 1st Qu.:  1.641  1st Qu.:  1.744  1st Qu.: 37.73  1st Qu.:  4.311
## Median :  2.876  Median :  3.235  Median : 52.82  Median :  5.638
## Mean    :  3.174  Mean    :  4.197  Mean    : 56.59  Mean    :  7.381
## 3rd Qu.:  5.280  3rd Qu.:  5.681  3rd Qu.: 64.29  3rd Qu.:  8.171
## Max.    : 14.231  Max.    :33.885  Max.    :186.68  Max.    :34.007
```

Figure 2: The histograms below show the distribution of GDP Growth and Unemployment Rates for all countries in the data set from 2004 to 2023. The density curves in red and blue provide a picture of the data's distribution. It's apparent that GDP growth is somewhat right skewed toward low positive growth rates. As for Unemployment rates, the data has a more bimodal distribution which is visible through the two distinct peaks in the density curve. The bimodal distribution makes sense given the data set includes developed and developing countries. The histograms serve as an aid to identify trends and possible outliers within the data.

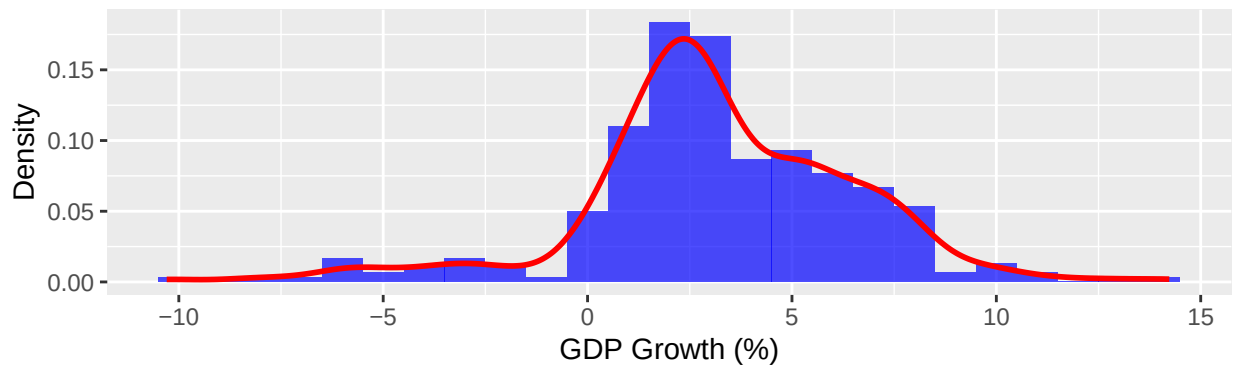
```
# Histogram for GDP Growth with Fitted Density
p1 <- ggplot(df, aes(x = GDP_Growth)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 1, fill = "blue", alpha = 0.7) +
  geom_density(color = "red", size = 1) + # Add fitted density curve
labs(title = "GDP Growth Distribution with Fitted Density",
      x = "GDP Growth (%)",
      y = "Density")
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
# Histogram for Unemployment with Fitted Density
p2 <- ggplot(df, aes(x = Unemployment)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 1, fill = "red", alpha = 0.7) +
  geom_density(color = "blue", size = 1) + # Add fitted density curve
labs(title = "Unemployment Rate Distribution with Fitted Density",
      x = "Unemployment Rate (%)",
      y = "Density")

# Arrange the plots vertically
grid.arrange(p1, p2, ncol = 1)
```

GDP Growth Distribution with Fitted Density



Unemployment Rate Distribution with Fitted Density

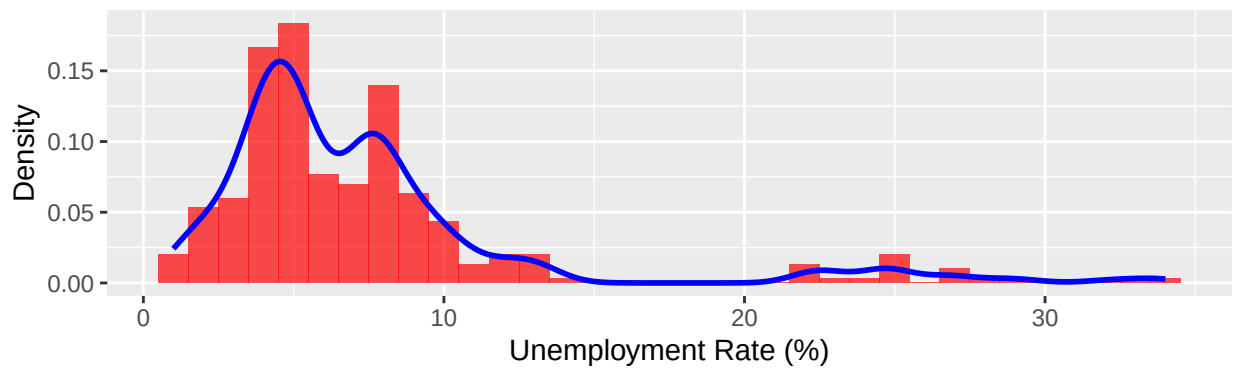


Figure 3: The boxplots below show the distribution of GDP Growth and Unemployment Rates for all countries in the data set from 2004 to 2023. The lines within the boxes display the median values and the IQR is displayed by the boxes themselves. Looking at the box plots it is evident that there are several outliers within the data set, especially for GDP Growth which has substantial outliers in both directions. The outliers bring attention to the fact that some of the countries experience economic recessions and booms, with other countries facing abnormally high unemployment rates at times.

```
# Boxplot for GDP Growth
p1 <- ggplot(df, aes(y = GDP_Growth)) +
  geom_boxplot(fill = "blue", alpha = 0.5) +
  labs(title = "Boxplot of GDP Growth", y = "GDP Growth (%)")

# Boxplot for Unemployment
p2 <- ggplot(df, aes(y = Unemployment)) +
  geom_boxplot(fill = "red", alpha = 0.5) +
  labs(title = "Boxplot of Unemployment Rate", y = "Unemployment Rate (%)")

# Stacks plots vertically
grid.arrange(p1, p2, ncol = 1)
```

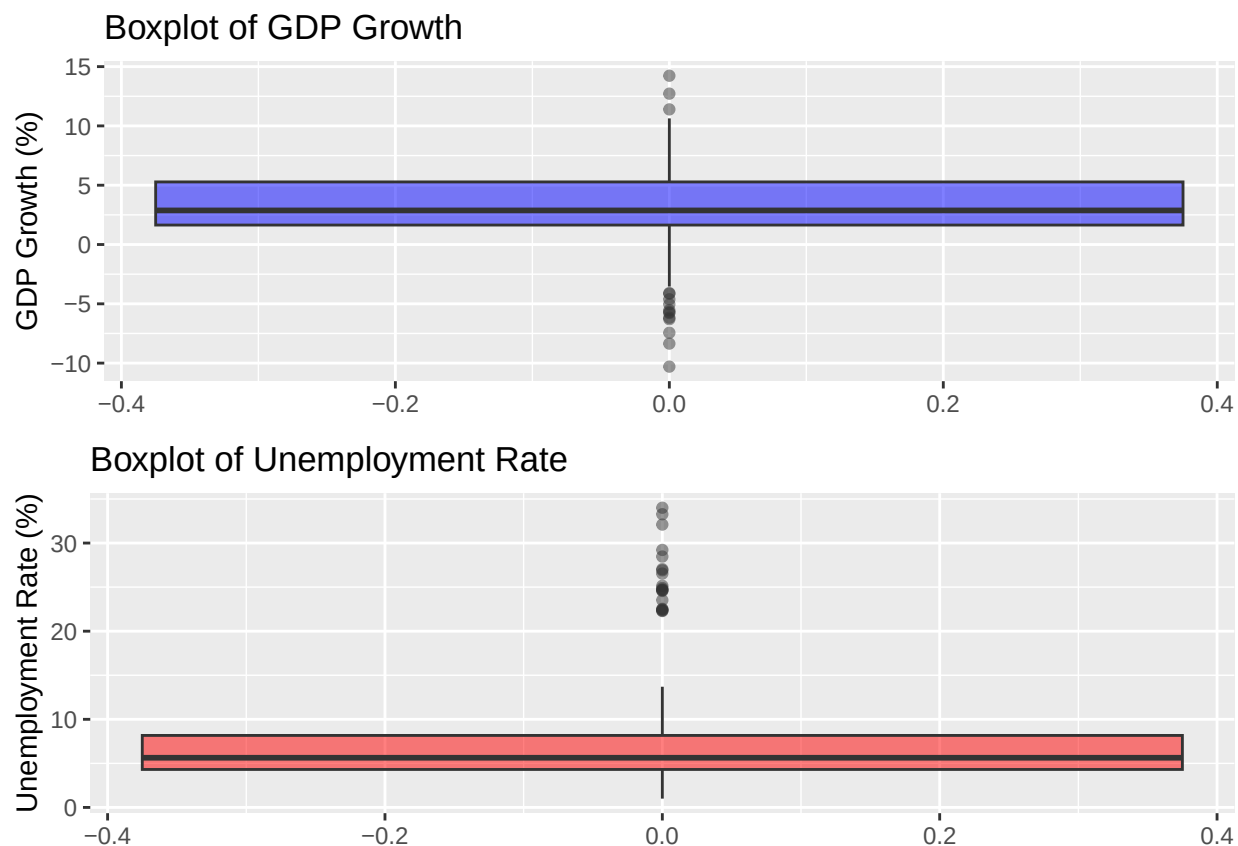


Figure 4: The correlation map below shows the relationships between GDP Growth, Unemployment, Inflation, and Trade Openness. It can be concluded that more economic growth is somewhat associated with lower levels of unemployment as evidenced by the weak negative correlation between GDP Growth and Unemployment in the correlation map. It's possible that Inflation and Trade Openness have limited influence since they appear weakly correlated with the other variables.

```
library(corrplot) # Load library for correlation plots

# Compute correlation matrix
cor_matrix <- cor(df[, c("GDP_Growth", "Unemployment", "Inflation", "Trade_Openness")], use = "complete")

# Plot correlation matrix
corrplot(cor_matrix, method = "color", type = "upper", tl.col = "black", tl.cex = 0.8)
```

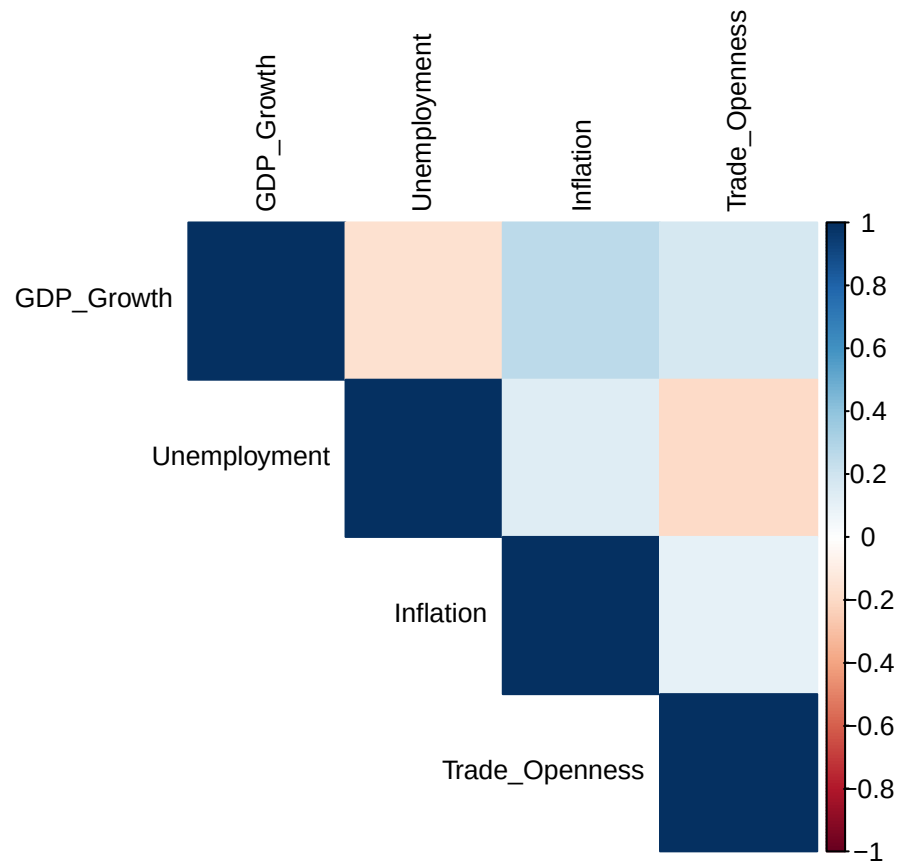


Figure 5: The scatterplots below show the relationships between the economic variables within the data set. The first plot displays a weak negative relationship between GDP Growth and Unemployment which again suggests there is a correlation between a country's growing economy and decreasing unemployment. The middle plot reveals a weak positive correlation between GDP Growth and Inflation which indicates that a growing economy can stimulate price increases. Finally, the bottom chart reveals a weak negative correlation between Trade Openness and Unemployment which is indicative that a more open economy may result in lower levels of unemployment. These trends are all visible through the red lines in the charts. Given the trends are weak, there is reason to believe that unknown factors are also influencing the relationships.

```
# Load necessary libraries
library(ggplot2)
library(gridExtra)

# Scatterplot: GDP Growth vs. Unemployment
p1 <- ggplot(df, aes(x = GDP_Growth, y = Unemployment)) +
  geom_point(alpha = 0.6, color = "blue") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(title = "Scatterplot: GDP Growth vs. Unemployment",
       x = "GDP Growth (%)",
       y = "Unemployment Rate (%)")

# Scatterplot: GDP Growth vs. Inflation
p2 <- ggplot(df, aes(x = GDP_Growth, y = Inflation)) +
  geom_point(alpha = 0.6, color = "green") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(title = "Scatterplot: GDP Growth vs. Inflation",
       x = "GDP Growth (%)",
       y = "Inflation Rate (%)")

# Scatterplot: Trade Openness vs. Unemployment
p3 <- ggplot(df, aes(x = Trade_Openness, y = Unemployment)) +
  geom_point(alpha = 0.6, color = "purple") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(title = "Scatterplot: Trade Openness vs. Unemployment",
       x = "Trade Openness (% of GDP)",
       y = "Unemployment Rate (%)")

# Define layout matrix to force separate rows
layout <- rbind(c(1), c(2), c(3))

# Arrange all scatterplots in one output (stacked vertically)
grid.arrange(p1, p2, p3, layout_matrix = layout)

## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```

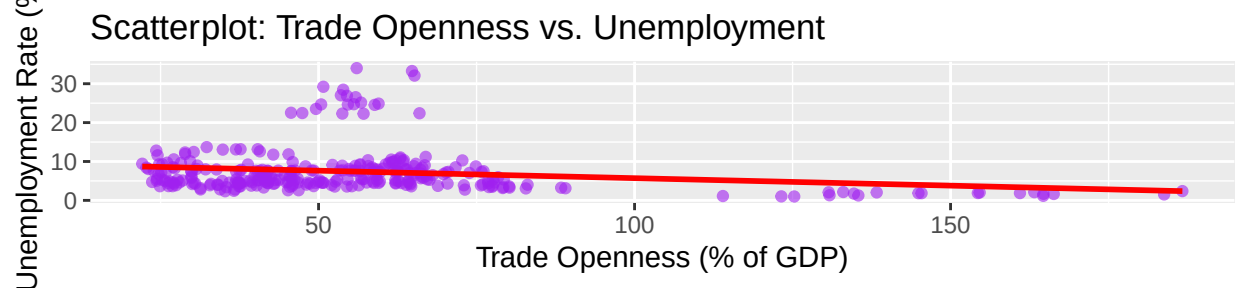
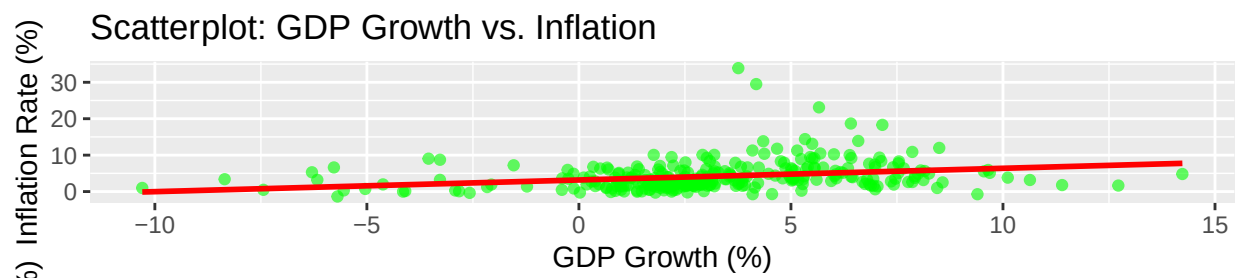


Figure 6: The line plots below display GDP Growth and Unemployment Rates for all countries within the data set from 2004 to 2023. There are visible fluctuations in the GDP Growth plot with declines in expected areas like the 2009 Global Financial Crisis and the 2020 COVID-19 pandemic. As for Unemployment Rates, the plot displays consistent rates for several countries whereas some have visibly high unemployment for extended periods of time. The plots are useful for gauging the impact that certain events have on global economies.

```
# Line plot for GDP Growth over time
p1 <- ggplot(df, aes(x = Year, y = GDP_Growth, group = Country, color = Country)) +
  geom_line() +
  labs(title = "GDP Growth Over Time", x = "Year", y = "GDP Growth (%)") +
  theme(legend.position = "right") # Keep legend on the right

# Line plot for Unemployment over time
p2 <- ggplot(df, aes(x = Year, y = Unemployment, group = Country, color = Country)) +
  geom_line() +
  labs(title = "Unemployment Rate Over Time", x = "Year", y = "Unemployment Rate (%)") +
  theme(legend.position = "right") # Keep legend on the right

# Combine the plots and share the legend
(p1 / p2) + plot_layout(guides = "collect") & theme(legend.position = "right")
```




```

# Convert Country to a factor for panel data indexing
df$Country <- as.factor(df$Country)

# Declare the panel data structure (Country and Year as index)
panel_data <- pdata.frame(df, index = c("Country", "Year"))

pooled_model <- plm(Unemployment ~ GDP_Growth + Inflation + Trade_Openness,
                    data = panel_data,
                    model = "pooling")

summary(pooled_model)

```

```

## Pooling Model
##
## Call:
## plm(formula = Unemployment ~ GDP_Growth + Inflation + Trade_Openness,
##      data = panel_data, model = "pooling")
##
## Balanced Panel: n = 15, T = 20, N = 300
##
## Residuals:
##      Min.   1st Qu.   Median   3rd Qu.    Max.
## -8.75793 -2.90332 -1.26041  0.84786 27.03937
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## (Intercept)   9.238488   0.751399 12.2950 < 2.2e-16 ***
## GDP_Growth   -0.308370   0.099720 -3.0924 0.0021750 **
## Inflation     0.279768   0.081027  3.4528 0.0006358 ***
## Trade_Openness -0.036282   0.011038 -3.2871 0.0011344 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    9794.5
## Residual Sum of Squares: 8898.4
## R-Squared:              0.09149
## Adj. R-Squared: 0.082282
## F-statistic: 9.9361 on 3 and 296 DF, p-value: 2.9239e-06

```

```
fixed_model <- plm(Unemployment ~ GDP_Growth + Inflation + Trade_Openness,
                  data = panel_data,
                  model = "within")

summary(fixed_model)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = Unemployment ~ GDP_Growth + Inflation + Trade_Openness,
##      data = panel_data, model = "within")
##
## Balanced Panel: n = 15, T = 20, N = 300
##
## Residuals:
##      Min.    1st Qu.      Median    3rd Qu.      Max.
## -3.422940 -0.990435 -0.049629  0.659725  8.242564
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## GDP_Growth    -0.1173677  0.0398105  -2.9482 0.003464 **
## Inflation     -0.0348199  0.0339904  -1.0244 0.306522
## Trade_Openness  0.0048982  0.0119261   0.4107 0.681599
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    829.98
## Residual Sum of Squares: 800.6
## R-Squared:    0.035407
## Adj. R-Squared: -0.022743
## F-statistic: 3.4504 on 3 and 282 DF, p-value: 0.017068
```

```

random_model <- plm(Unemployment ~ GDP_Growth + Inflation + Trade_Openness,
                    data = panel_data,
                    model = "random")

summary(random_model)

```

```

## Oneway (individual) effect Random Effect Model
##   (Swamy-Arora's transformation)
##
## Call:
## plm(formula = Unemployment ~ GDP_Growth + Inflation + Trade_Openness,
##      data = panel_data, model = "random")
##
## Balanced Panel: n = 15, T = 20, N = 300
##
## Effects:
##              var std.dev share
## idiosyncratic  2.839   1.685 0.079
## individual    32.878   5.734 0.921
## theta: 0.9344
##
## Residuals:
##      Min.   1st Qu.   Median   3rd Qu.    Max.
## -3.04793 -0.97313 -0.17519  0.59228  9.46536
##
## Coefficients:
##              Estimate Std. Error z-value Pr(>|z|)
## (Intercept)    7.7317055   1.6200359   4.7726 1.819e-06 ***
## GDP_Growth    -0.1176059   0.0397092  -2.9617  0.00306 **
## Inflation     -0.0322214   0.0338987  -0.9505  0.34185
## Trade_Openness  0.0027841   0.0116332   0.2393  0.81085
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    868.52
## Residual Sum of Squares: 839.17
## R-Squared:    0.033791
## Adj. R-Squared: 0.023999
## Chisq: 10.3521 on 3 DF, p-value: 0.015799

```

```
hausman_test <- phptest(fixed_model, random_model)
hausman_test
```

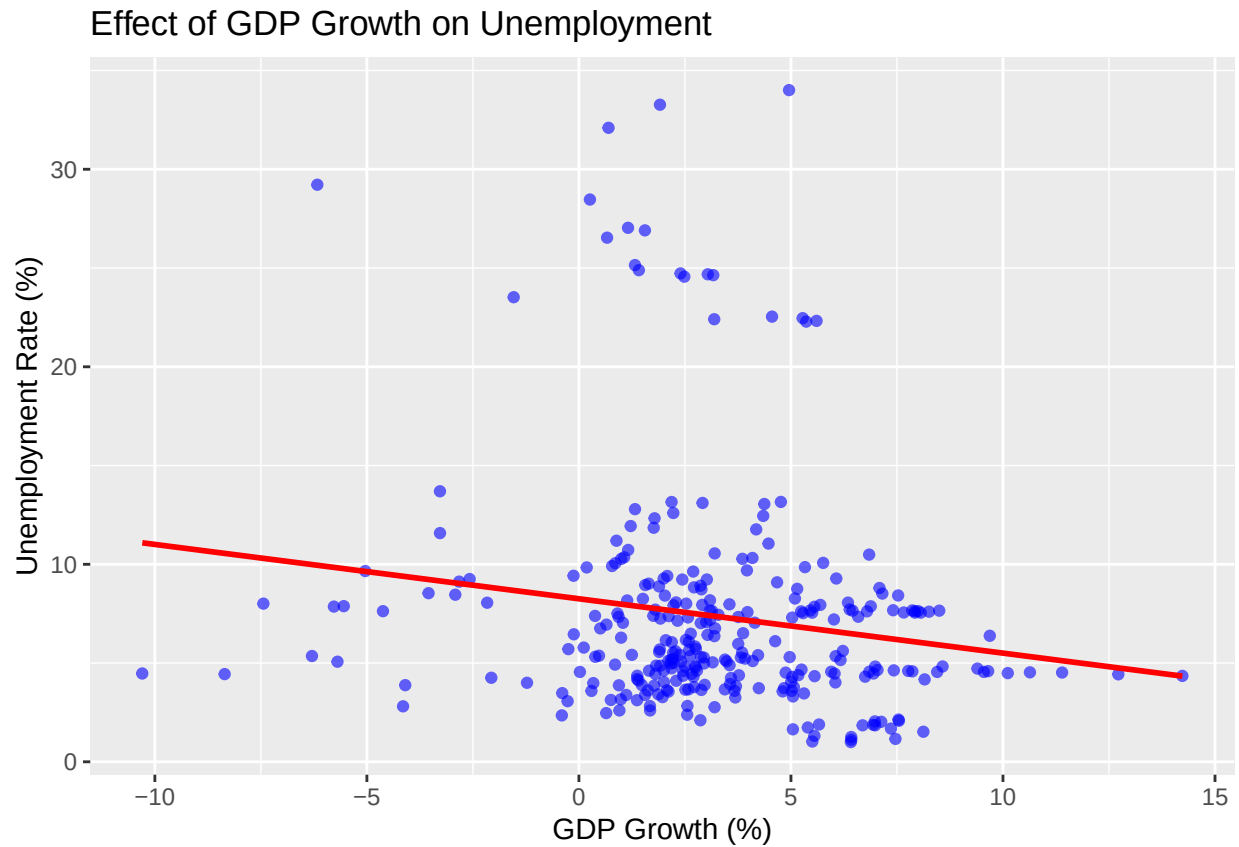
```
##
## Hausman Test
##
## data: Unemployment ~ GDP_Growth + Inflation + Trade_Openness
## chisq = 1.8846, df = 3, p-value = 0.5967
## alternative hypothesis: one model is inconsistent
```

```
stargazer(pooled_model, fixed_model, random_model, type = "text")
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               Unemployment
##                               (1)          (2)          (3)
## -----
## GDP_Growth          -0.308***          -0.117***          -0.118***
##                      (0.100)          (0.040)          (0.040)
##
## Inflation           0.280***           -0.035           -0.032
##                      (0.081)          (0.034)          (0.034)
##
## Trade_Openness      -0.036***           0.005           0.003
##                      (0.011)          (0.012)          (0.012)
##
## Constant            9.238***              7.732***
##                      (0.751)              (1.620)
## -----
## Observations           300              300              300
## R2                     0.091              0.035              0.034
## Adjusted R2            0.082             -0.023              0.024
## F Statistic    9.936*** (df = 3; 296) 3.450** (df = 3; 282) 10.352**
## =====
## Note:                               *p<0.1; **p<0.05; ***p<0.01
```

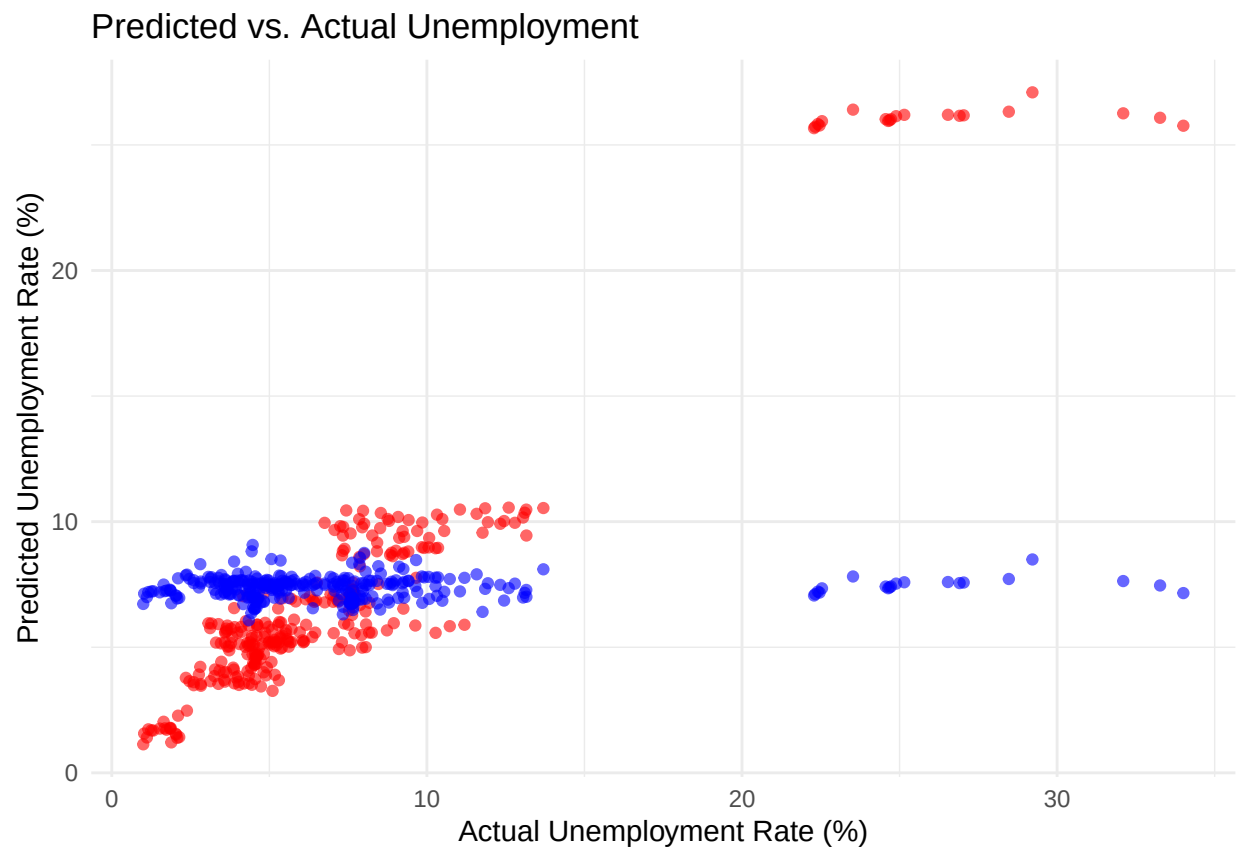
```
ggplot(df, aes(x = GDP_Growth, y = Unemployment)) +
  geom_point(alpha = 0.6, color = "blue") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(title = "Effect of GDP Growth on Unemployment",
       x = "GDP Growth (%)",
       y = "Unemployment Rate (%)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
df$pred_fixed <- predict(fixed_model)
df$pred_random <- predict(random_model)

ggplot(df, aes(x = Unemployment)) +
  geom_point(aes(y = pred_fixed), color = "red", alpha = 0.6) +
  geom_point(aes(y = pred_random), color = "blue", alpha = 0.6) +
  labs(title = "Predicted vs. Actual Unemployment",
       x = "Actual Unemployment Rate (%)",
       y = "Predicted Unemployment Rate (%)") +
  theme_minimal()
```



Panel Data Model Estimation and Selection

The analysis seeks to assess the relationship between GDP and unemployment using a panel data set that encompasses 14 different developed and developing countries. The analysis also uses control variables such as inflation and trade openness that possibly play a role in unemployment as well. The Pooled OLS model, Fixed Effects model, and Random Effects model were estimated and assessed for which is the best fit. The Pooled OLS model does show that GDP growth is negatively correlated with unemployment. As for inflation and trade openness, inflation is positively correlated with unemployment and trade openness is negatively correlated with unemployment. The issue with the model is that it has a low R^2 of 0.091 meaning it only explains 9% of the variation in the data. Another big issue with the model is that it does not account for country specific effects which leaves it susceptible to omitted variable bias. The Fixed Effects model removes between country variation which controls for the unobserved country specific factors. Compared to the Pooled OLS model, GDP is still statistically significant but less so. In this model inflation and trade openness are now insignificant which indicates that their influence is likely due to differences across countries. The R^2 of 0.035 is even lower than in the pooled model, indicating a weaker overall fit. The Random Effects model delivered very similar results to the Fixed Effects model, with GDP remaining statistically significant, while inflation and trade openness are insignificant. The R^2 is 0.034, almost identical to the Fixed Effects model.

The performance of the Hausman test yielded a p-value of 0.5967 which is not statistically significant. This result implies that the Random Effects model is the better option. Also, the plot comparing Predicted vs Actual Unemployment Rates reveals the Random Effects model offers a reasonable fit given its predicted values cluster around actual unemployment rates. On the other hand, the Fixed Effect mode doesn't have the same accuracy of predictions. Given that the Random Effects model accounts for country specific effects while preserving between country variation, it is the best option for analyzing the relationship between GDP growth and unemployment. The results show that there is a consistent negative relationship between GDP growth and unemployment. Additionally, once country specific effects are considered, inflation and trade openness don't appear to significantly impact unemployment which is likely due to differing economies rather than a large scale trend.